

INTERCEPTION AND RECHARGE PROCESSES BENEATH A *PINUS ELLIOTII* FOREST

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ABSTRACT

Interception and recharge processes beneath a *Pinus elliotii* forest were considered in an integrated study. In the study area, annual rainfall was divided into throughfall (74.45%), stemflow (9.37%) and interception (16.28%). Throughfall and stemflow infiltrate into the soil in different ways. The results show that trees can affect the recharge characteristics by providing throughfall as a non-point source and stemflow as a point source, and also through their influence on infiltration processes by making the hydraulic conductivity of soil heterogeneous. In the root zone there was a divergent zero flux plane recharged by macropore flow during heavy rain and a convergent zero flux plane caused by transpiration during dry periods.

KEY WORDS forest interception; throughfall; stemflow; soil water movement; zero flux plane

INTRODUCTION

From the viewpoint of water input to a forested basin, how rainfall is intercepted by trees and how throughfall and stemflow infiltrate into the soil are the key processes. The role of interception and the associated redistribution of rainfall in water resource management has been examined in both temperate (Rothacher, 1963; Rutter, 1963; Helvey and Patric, 1965; Clements, 1971) and tropical ecosystems (Jackson, 1975). Catchment recharge by rainfall will be very different in forested and unforested areas, and will be related to the type of trees, e.g. mountain beech forest (Rowe, 1975), pine forests (Schulze *et al.*, 1978; Pook *et al.*, 1991), eucalypt forests (Pook *et al.*, 1991a, b), cocoa plantations (Miranda, 1994), black locust plantation (Wang, 1992), semi-deciduous forest groves (San Jose and Montes, 1992) and beech coppice (Giacomin and Trucchi, 1992).

There are two processes by which rainwater infiltrates into the ground: soil matrix flow and macropore or pipe flow. In the former, the theory of potential flow can be used by assuming that all the soil moisture continuously participates in the flow process, that water storage and water flow are strongly related to one another at the microscopic scale and that potential energy consumption during flow is a diffusive process. However, macropore or pipe flow is likely to be important in structured soils, such as cracked clay soils, or where root holes or other biopores exist in abundance. Beven and Germann (1982) have summarized the hydrological implications of macropores and White (1985) provides a more pedological and agronomic perspective. Kirkby (1985) presented a concept of pipe flow at the hillslope scale and clearly established its importance. Mosley (1978) emphasized the importance of pipe flow in a forested catchment in New Zealand.

Although the natural recharge of forested catchment depends largely on the pattern of throughfall and stemflow and soil conditions in the forest, most workers have focused only on interception loss and rainfall redistribution above the forest floor, rather than the integrated relationship between net rainfall and infiltration.

In this paper, both interception and recharge processes within pine plantations will be discussed in detail.

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EXPERIMENTAL SITE

Hydrological observations have been conducted for two years (since 1990) at the experimental site on a small hill on the campus of Zhongshan University, Guangzhou, China (113° 30' E, 23° 28' N). It has an average annual temperature of 21.8°C and an average annual precipitation of 1500 mm. The slope of the hill is about 6°. This supports *Pinus elliotii* plantations with an average stand density of 400 trees/ha. The average height of the pines is about 20 m and the average diameter about 30 cm. Three automatic rain gauges monitor precipitation, throughfall and stemflow. To ensure representative throughfall measurements, five pans (30 cm in diameter) were placed randomly within the study area and throughfall was measured once a day. Additionally, spiral stemflow gauges were used to measure water flowing down the trunks of five randomly selected trees. The gauges were constructed from 100 cm long rubber tubing fixed around the trunk with wires and sealed to the bark with silicone rubber caulking. The spiral collectors encircled the trunks about 120 cm above the ground. The stemflow was collected in plastic drums and measured once a day. The daily variations of throughfall and stemflow were used to correct the data provided by the automatic rain gauges.

Soil water and water-table levels were measured with tensiometers at depths of 20, 50, 100, 150 and 200 cm, and a piezometer at a depth of 300 cm. The observations of the tensiometers and the piezometer were undertaken once or twice a day. The total potential of the vertical profile of the soil was calculated by setting the surface as the reference.

The soil in the experimental site was a typical lateritic red earth with a bulk density of 1.32 g/cm³, which was characterized by loss of silica and enrichment of aluminium under the conditions of high temperature and high humidity in this subtropical region. With a lack of organic matter and a high acidity (mean pH 4.55), this lateritic red earth is mainly composed of sand (16.19%), silt (54.14%) and clay (29.67%). The hydraulic conductivity varies markedly in space, but tended to decrease with distance from the trees, around which many cracks were found. The saturated hydraulic conductivity of the lateritic red earth without cracks was about 4.5×10^{-4} cm/s. Water-table levels were normally 2–3 m deep.

WATER BALANCE

Precipitation was partitioned into the loss associated with the tree canopy (T') and stems (S'), which dry after rainfall stops, throughfall (T), stemflow (S) and evaporation E . Therefore, the water balance in the forest can be expressed as

$$P = T + S + T' + S' + E$$

or

$$P = T + S + I$$

where I is the total loss ($= T' + S' + E$). Saturated at the end of rainfall, T' and S' approximate the canopy storage capacity and stem storage capacity, respectively. Based on field data, the correlations between precipitation and throughfall and between precipitation and stemflow were fairly high ($r = 0.91$ and 0.80 , respectively) (Figure 1). Throughfall is 74.45% of the gross precipitation and stemflow is 9.37%. The relationships between precipitation, throughfall and stemflow can be described statistically with two linear regression functions

$$T = 0.904(P - 1.6)$$

$$S = 0.088(P - 4.32)$$

The variations of throughfall, stemflow and interception with rainfall at the experimental site are given in Table I. Throughfall and stemflow trend to increase with rainfall amount from 54.35 to 80.37% and 0.9 to 10.43%, respectively. In spite of the increase of interception volume with rainfall amount, its magnitude as a percentage rainfall declines rapidly from 44.71 (0–5 mm class) to 9.21% (> 25 mm class).

When the rainfall is low (less than 5 mm), stemflow is limited and interception plays an important part in

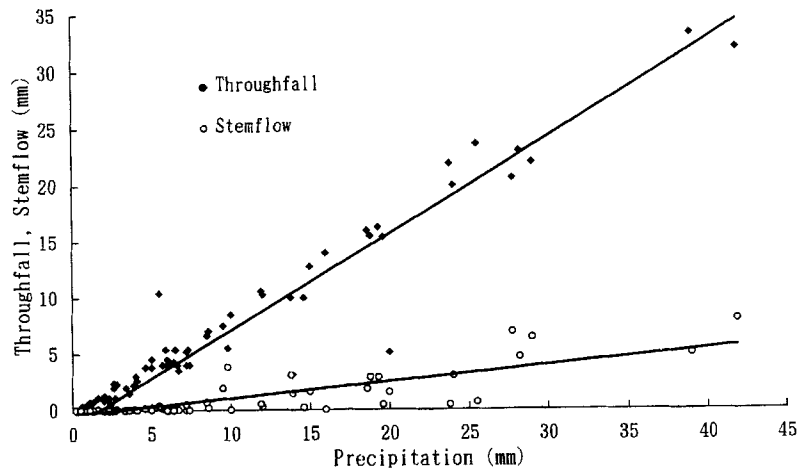


Figure 1. Relationship between amount of precipitation, throughfall and stemflow

rainfall partitioning. As a result, only about 55.29% of the incident rainfall is expected to reach the forest floor of the study area. For rainfall in the range 5–25 mm, the percentage partitioning between throughfall, stemflow and interception changes by no more than 5%. For rainfall events greater than 25 mm, throughfall and stemflow become increasingly important in rainfall partitioning and more than 90% of the incident rainfall can be observed beneath the forest.

Table II shows the seasonal distributions of rainfall and its partitioning in the experimental site from January 1990 to December 1991. In spring and summer, there were many rainfall events over 30 mm. More than 76 and 9% of the rainfall was accounted for by throughfall and stemflow, respectively. In autumn and winter most rainfall events were less than 15 mm and interception became more important than in spring or summer. During rainfall events over the year, about 83.82% of the annual precipitation is available for recharge to the experimental site.

PROCESSES OF PARTITIONING PRECIPITATION DURING EVENTS

Figure 2 shows the variation of the intensities of precipitation, throughfall and stemflow on 31 July 1990, a rainfall event which was preceded by seven days with no rain. The total precipitation was 27.8 mm and it began at 02:00, with throughfall starting 25 minutes later. At this time, the precipitation input had reached 1.5 mm and this could be considered as the canopy capacity. Once the canopy capacity was reached, throughfall responded promptly to rainfall. Stemflow of 0.01 mm/h was detected after 60 minutes, when precipitation and throughfall had reached 3.7 and 1.7 mm, respectively. After 230 minutes from the start

Table I. Variation of throughfall, stemflow and interception with amount of rainfall at the study area

Rainfall (mm)	Throughfall (%)	Stemflow (%)	Interception (%)
0–5	54.35	0.94	44.71
5–10	69.32	7.33	23.35
10–15	71.71	5.58	22.71
15–20	71.60	7.87	20.53
20–25	73.88	7.99	18.13
> 25	80.37	10.42	9.21

Table II. Seasonal distribution of rainfall and its partitioning in the study area

Season	Throughfall (%)	Stemflow (%)	Interception (%)
Spring (February–April)	77.73	9.30	12.97
Summer (May–July)	76.09	9.08	14.83
Autumn (August–October)	63.90	13.78	22.32
Winter (November–January)	65.44	11.03	23.53
Year	74.45	9.37	16.18

of precipitation, stemflow increased markedly and reached 2.0 mm/h at 305 minutes. Because stem storage had been filled, the stemflow remained more steady than throughfall from 305 to 360 minutes.

The occurrence of throughfall depends largely on the saturation of the canopy. Once the canopy storage had been exceeded, throughfall occurred. The time lag between rainfall and throughfall becomes shorter if there is a previous event. Likewise, once stem storage was exceeded, stemflow could be expected in the experimental site. In fact, because of wind and the structure of the stem surfaces, the stems were not saturated from top to bottom over all the surface, but were saturated first from top to bottom on the windward side of the stem surface. Stemflow started suddenly and moved down the stem as a channel flow along the saturated stem surface rather than as a thin flow over all the stem. With increased precipitation, the saturated area extended over the whole surface of stem. As a result, the stemflow increased sharply.

After precipitation had stopped, some intercepted rainfall would fall as throughfall, especially on windy days, or flow down as stemflow. Therefore, the end-points for precipitation, throughfall and stemflow followed the rule of first in, first out.

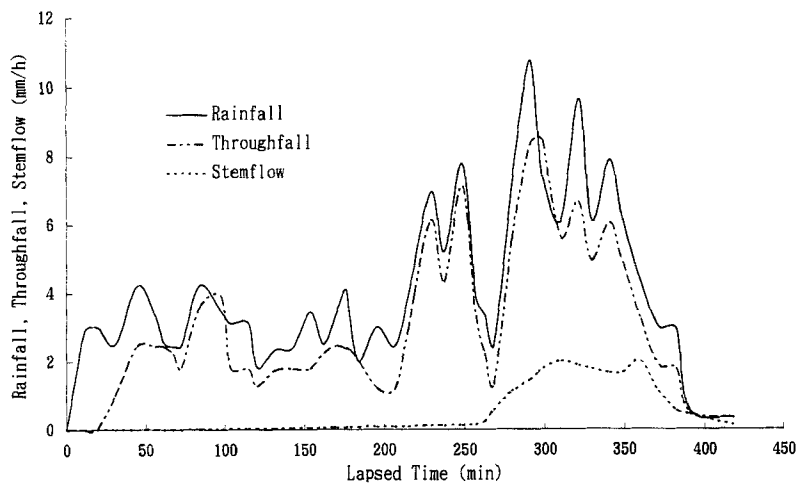


Figure 2. Variation of rainfall, throughfall and stemflow from 02:00 to 09:00 on 31 July 1990

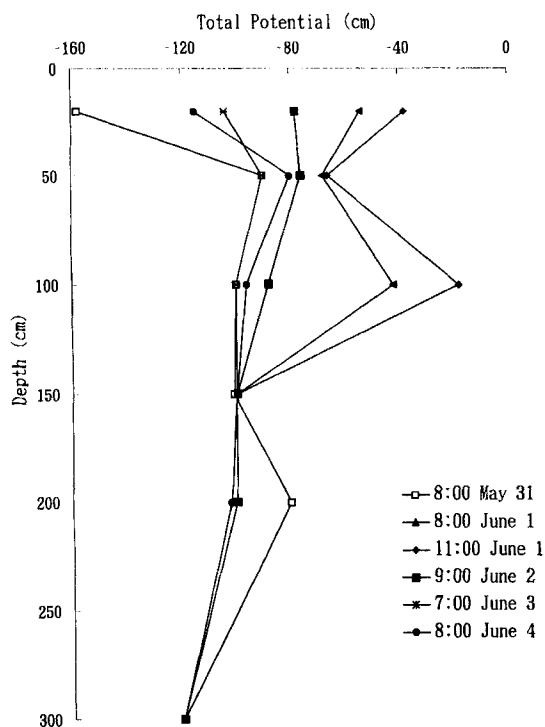


Figure 3. Vertical profiles of total potential at the study site

PROCESS OF WATER ENTERING THE GROUND

In the forested area, the distribution of tree roots has a great effect on the water content of the soil as a result of both take up of water and pipes. In the root zone, the zero flux plane plays an important part in controlling water movement in the ground. Figure 3 shows the vertical profile of total potential variations at the experimental site. The period of rain extended from about 18:00 to 23:00 on 31 May and for a whole day on 1 June 1990. It was found that there were both convergent and divergent zero flux planes at different depths. In particular, the variations of total potential in the top 100 cm of the soil were rapid. There was a divergent and a convergent zero flux plane in the 50–100 cm depth range during dry days and after rain, respectively. In the top 50 cm, the soil water moved upward in response to evaporation and downward as infiltration before and after rainfall, respectively. Water movement in the 50–150 cm layer was influenced by the many tree roots. These would absorb water as transpiration, thus causing a convergent zone for water in the vertical profile. During rainfall, the roots would act as a pipe enabling water to enter the ground rapidly, thus promoting recharge directly to the root zone and establishing the divergent zero flux plane. In this way, evapotranspiration could be calculated for each zone bounded by zero flux planes. Rain water infiltrating as matrix flow provided another way for water to enter the soil. The infiltrating water was mainly contributed by throughfall. It was also found that infiltration was confined to the top of the soil during rainfall.

The occurrence of a large water potential drop at the interface of the soil and root surface with rapidly transpiring plants is generally accepted. It has been assumed that this gradient is attributable to a reduced area of contact between water in the soil and roots. Therefore, the existence and distribution of the zero flux plane can control the direction of unsaturated infiltration. Zero flux planes in forest soils reflect the influence of roots, infiltration and evaporation. Tree roots not only help rain water to infiltrate rapidly into the soil, but also establish a zero flux plane, where the soil water changes greatly in amount and direction through space and time.

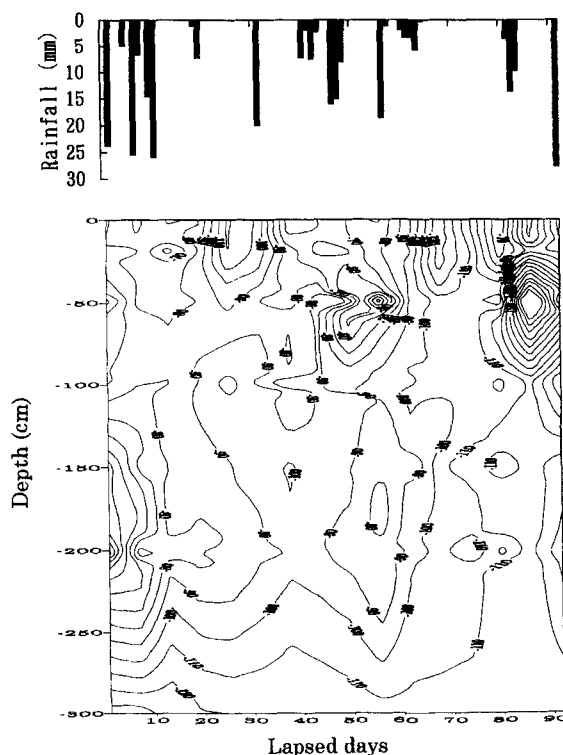


Figure 4. Daily variation of rainfall and total potential from 1 May 1990 to 31 July 1990

Figure 4 shows the daily variation of rainfall and total potential in the soil during summer (from 1 May to 31 July 1990). During this period, there were many rainfall events and the total potential in the soil was high. During rainfall events, the total potential in the vertical profile shows that the water content in the topsoil (less than 50 cm) and in the soil at 100 cm depth increased rapidly. During periods of no rainfall water was quickly lost from the 100 cm depth soil. As a whole, the soil water tends to move downwards.

Figure 5 shows the daily variation of rainfall and total potential in the soil during winter (from 1 November 1990 to 31 January 1991). There were less rainfall events and the soil water content was low. From Figure 5, the convergent zero flux plane was found in the soil at 100 cm depth. It seems that the soil water was taken up by the roots. At the same time, a divergent zero flux plane was evident in the soil at 200 cm depth. During the rainfall event, the total potential responded quickly in the topsoil (less than 50 cm) and in the soil 100 cm depth.

CONCLUSIONS

Figure 6 is a conceptual diagram to show how rainfall is partitioned during a rainfall event. Interception loss can also be considered as evaporation. The remaining water is divided into throughfall and stemflow. The importance of throughfall and stemflow is reflected not only in their amounts, but also in the variations they portrayed in space and time. Stemflow tended to increase with precipitating amount, particularly when the latter was small. Interception decreased with precipitation amount, but showed an increase for high rainfall amounts. Throughfall showed an opposite trend to interception. If the period of rainfall became longer, and the precipitation intensity was small, throughfall was small and interception remained constant. The intercepted water could flow down rapidly as stemflow, because of saturation of the stem surfaces. On average, throughfall occurred about 20–40 minutes later and stemflow about 60–250 minutes later than the onset of rainfall in the experimental site.

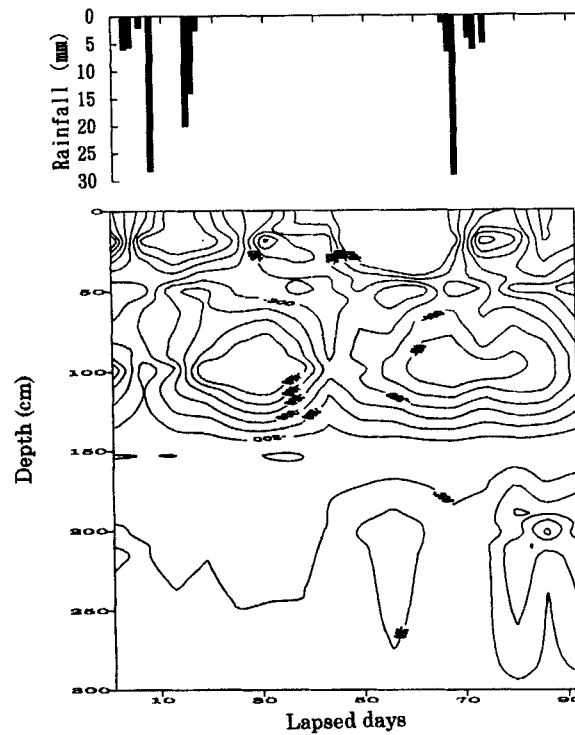


Figure 5. Daily variation of rainfall and total potential from 1 November 1990 to 31 January 1991

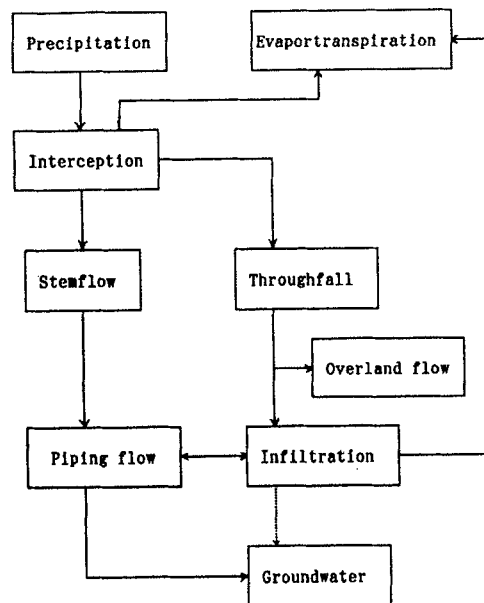


Figure 6. Conceptual diagram for partitioning of precipitation at the research site

The redistribution of precipitation and the heterogeneity of the soil hydraulic conductivity have a great effect on the processes of water entering the soil. In fact, throughfall could be considered as a non-point source over the whole catchment and stemflow as a point source for recharge to the catchment. Throughfall always occurred before stemflow and ranged from 50 to 75% of gross rainfall. The pattern of throughfall was similar to that of rainfall. Compared with throughfall, stemflow increased sharply and then became constant after stem storage had been filled.

Throughfall would also contribute to recharge as infiltration. This water eventually left the site as evapotranspiration and contributed less to groundwater. However, stemflow entered the ground directly through pipes. Field observations showed these pipes could transmit water rapidly into the ground, thus limiting the possibility of overland flow in the research area. Thus stemflow could be the major source for recharging groundwater, as it could reach greater depths along the pipes existing at the site.

From these analyses, the presence of a forest can be seen to cause the redistribution of precipitation and to result in heterogeneity of the hydraulic conductivity in the soil. Both exert important effects on variations of water recharge in space and time. In conclusion: (1) from the viewpoint of the water balance, throughfall and stemflow amounted to about 74% and 9% of gross precipitation; (2) recharge characteristics at the site were dependent on the partitioning of precipitation and spatial variations in hydraulic conductivity beneath the forest; (3) throughfall can be considered as an non-point source and stemflow as a point source; and (4) rainwater was partitioned as throughfall, which infiltrated into the top layer of the soil and left as evapotranspiration, and as stemflow, which recharged groundwater via pipe flow around the roots of trees.

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